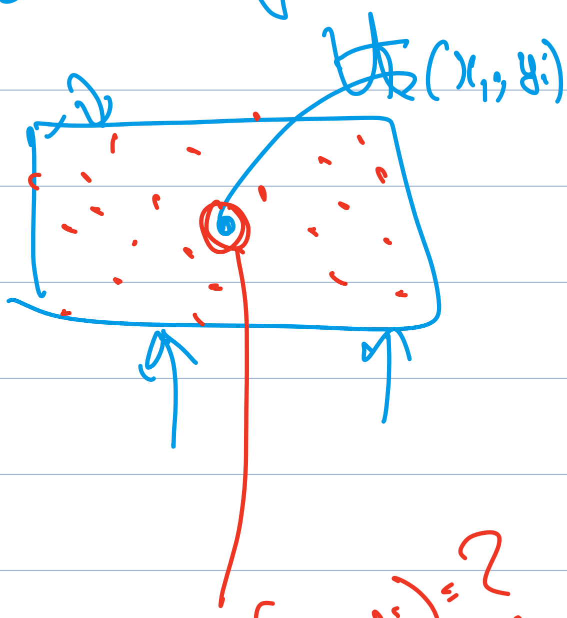
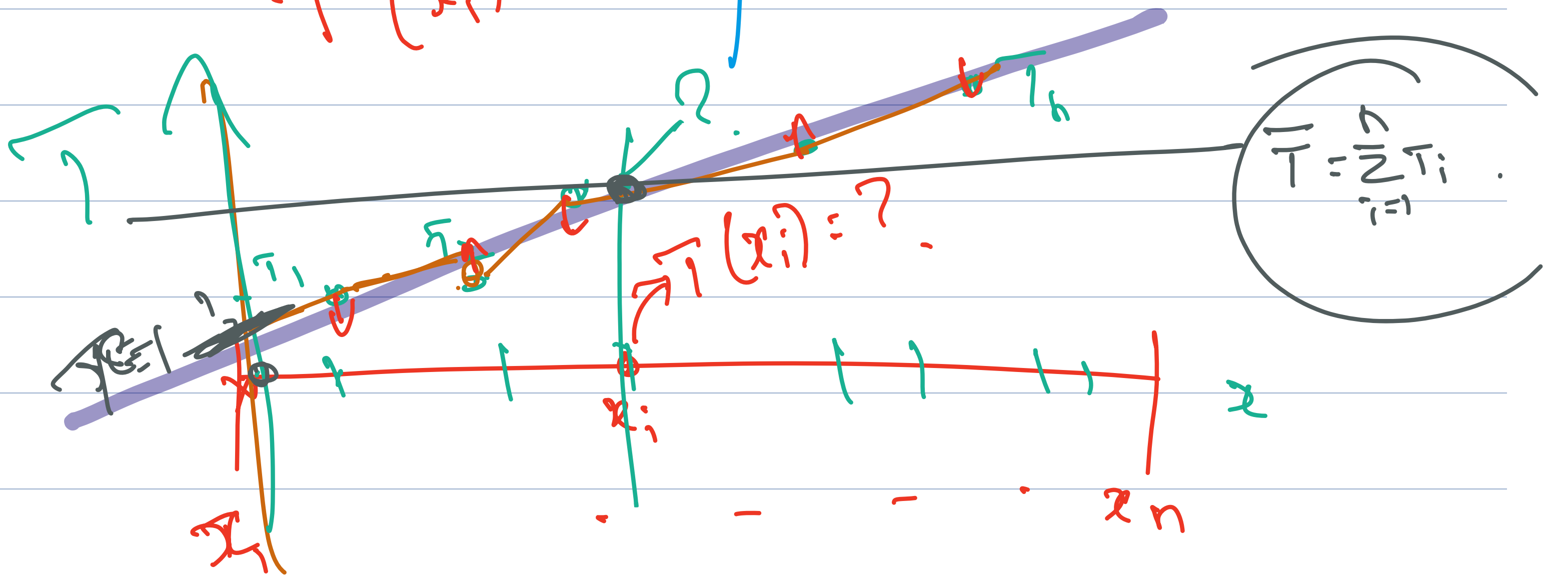
Poisson

$$\to \varepsilon \Delta u = f$$



$T(x_i, y_i) = ?$

(x_i, y_i)	T_i
$(0,0)_1$	T_1
$(1,0)_2$	T_2
$\vdots$	
$(\ )_n$	T_n



$$y(x) = mx + c$$
$$= wx + c$$
$$= wx + 1$$

When $x = 0$,
$y = c$

slope?

$y(x) = wx$ $\qquad\qquad (x_i, y_i),\ i = 1, \dots, n,$

let $w_0 = 0.1$

$y(x_1) = \tilde{y}_1 = w_0 x_1$

$y(x_2) = \tilde{y}_2 = w_0 x_2$

$\vdots$

$y(x_n) = \tilde{y}_n = w_0 x_n$

Total error

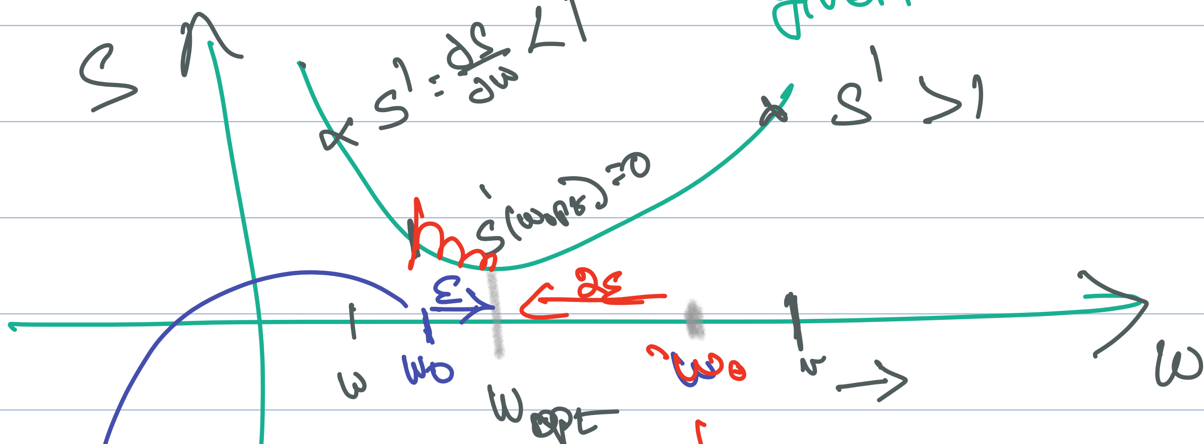
$$S(\cdot) = \sum_{i=1}^{n} \|\tilde{y}_i - y_i\|^2 = \sum_{i=1}^{n} (w_0 x_i - y_i)^2$$

suppose $w_{opt} = w_0 \Rightarrow S \to 0$

only variable

$$\min S(w) = \sum_{i=1}^{n} (w_0 x_i - y_i)^2$$

given



if $w^{opt} = w_0 + \varepsilon$;

else $w^{opt} = w_0 - 2\varepsilon$

$\varepsilon > 0$

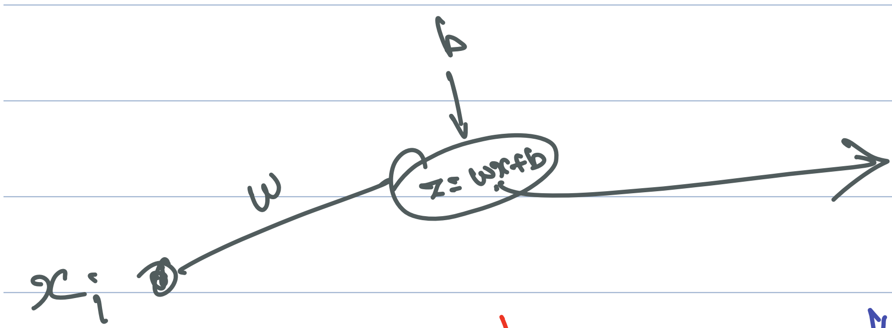
$$w^{opt} = w_0 - \text{Sign}\left(\frac{\partial S}{\partial w}\right) \cdot \varepsilon$$

$$w^{opt} = w_0 - \text{Sign}\left(\frac{\partial S}{\partial w}\right) \cdot \varepsilon$$

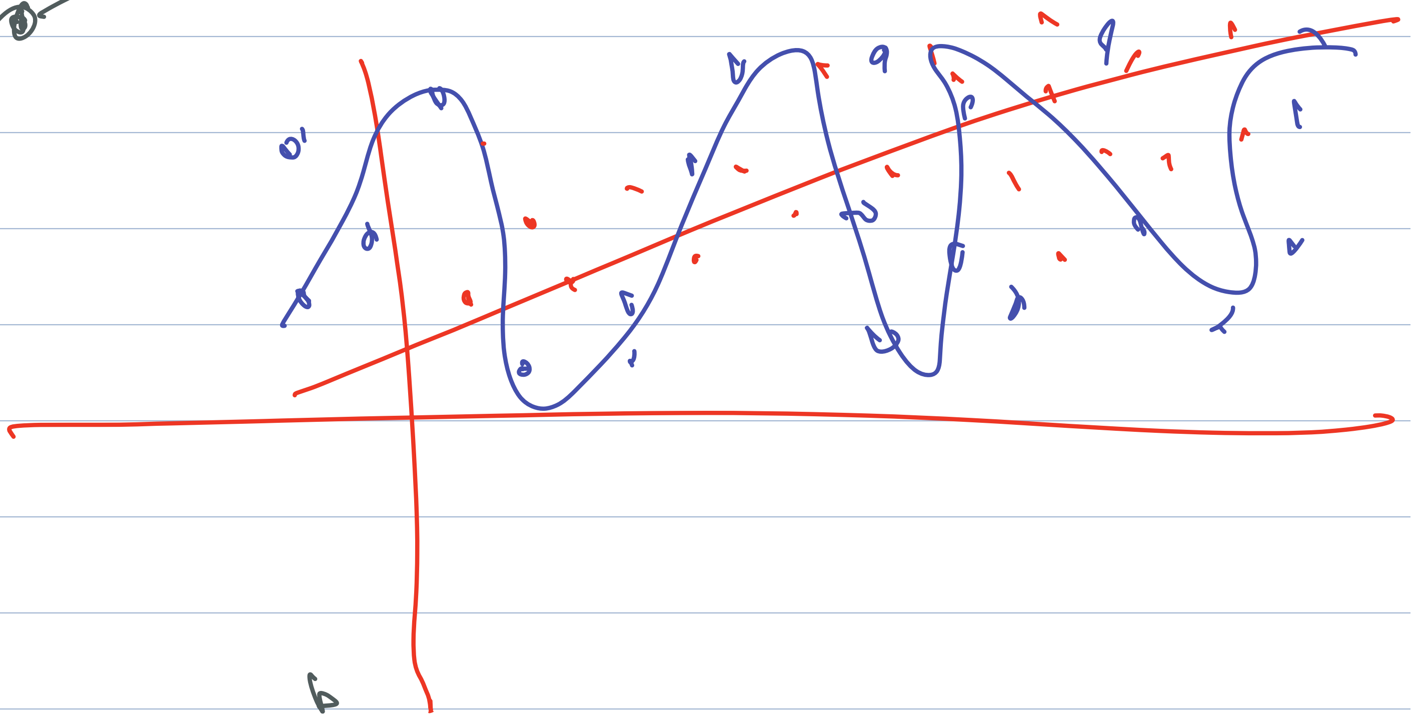
learning rate

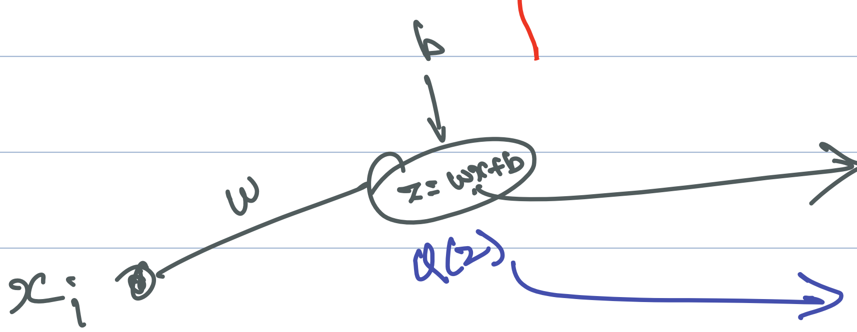
$$w^{opt} = w_0 - \cancel{\text{Sign}}\left(\frac{\partial S}{\partial w}\right) \cdot \eta$$

$$\boxed{w^{k+1} = w^{k} - \eta \nabla S}$$



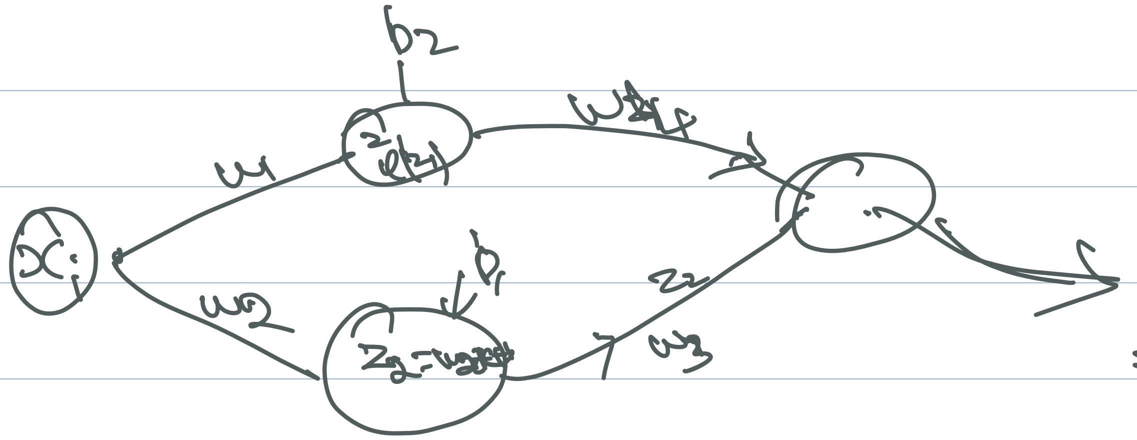
$\tilde{y}(x_i) = wx + b$. (S.L.R)



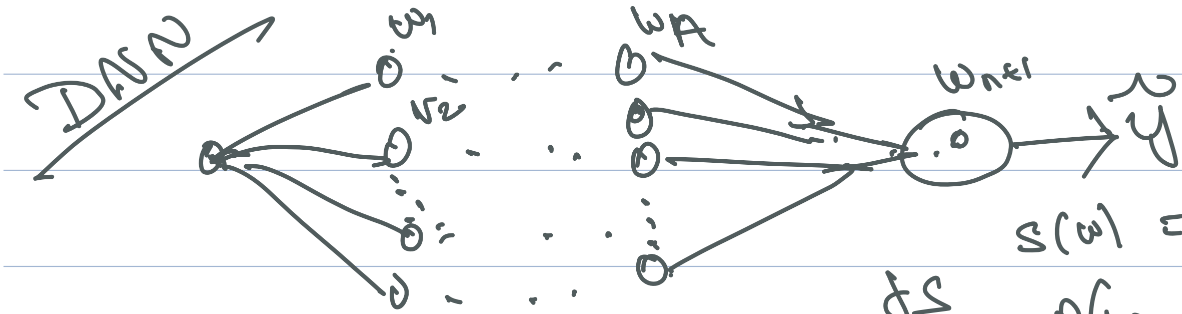


$\cancel{\tilde{y}(x_i)} = wx + b$. SLR

$\tilde{y} = \varphi(wx + b)$



$$\tilde{y} = \phi(w_4 z_1 + w_3 z_2 + b)$$

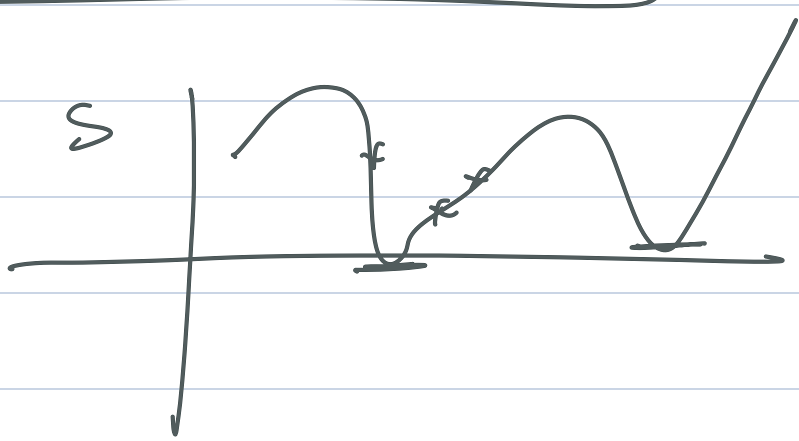


$$S(w) = \phi(w_{n+1} z_n + b + \cdot)$$

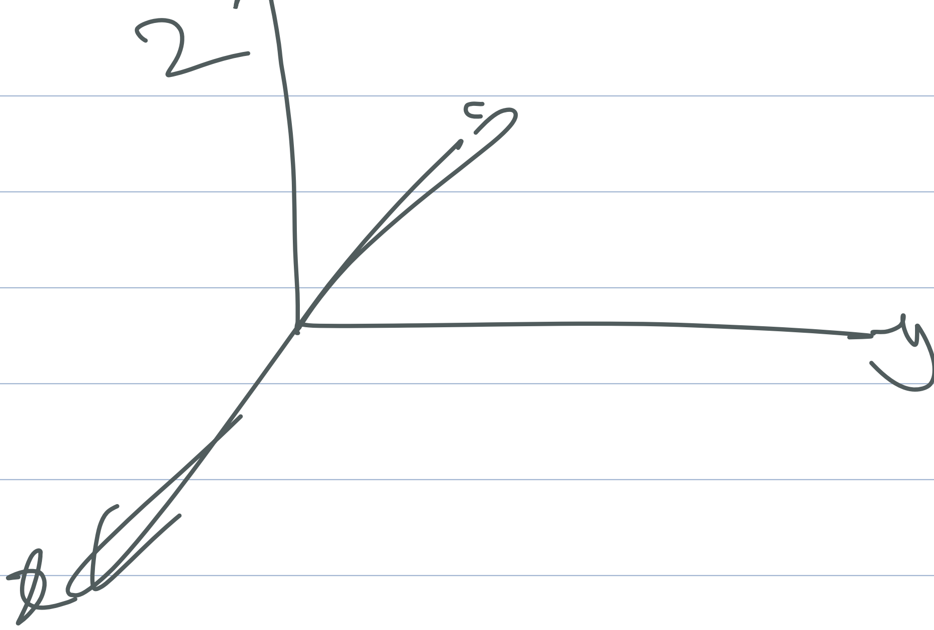
$$\frac{dS}{dw_n} = \phi(w_{n+n}(\phi(w_n z_{n-5} + b) + \dots$$

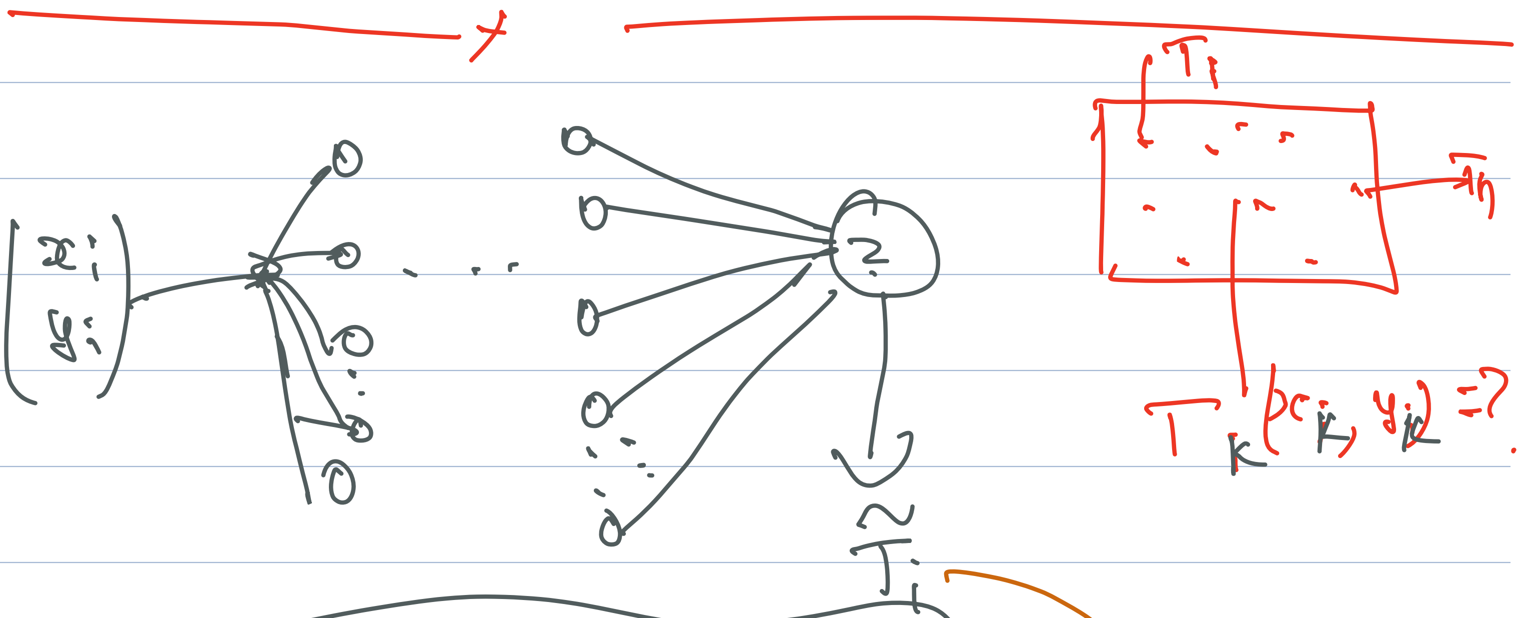
$$\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}^{k+1} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}^{k} - \eta \begin{pmatrix} \frac{dS}{dw_1} \\ \vdots \\ \frac{dS}{dw_n} \end{pmatrix}$$

$$\boxed{W^{k+1} = W^{K} - \eta \nabla S}$$ ← Back Propagation



M





$$\min \; S(w) = \sum_{i=1}^{n} (\tilde{T}_i - T_i)^2$$

known

Poisson

$$-\varepsilon \Delta T = f$$

$$-\varepsilon \Delta T - f = 0$$

$$S(w) = \sum_{i=1}^{n} \left(-\varepsilon \Delta T_i(w) - f_i\right)^2$$

$\frac{\partial S}{\partial w}$

Physics Informed Neural Network

SciML field.

NSE

$$\frac{\partial u}{\partial t} - \frac{1}{Re}\Delta u - \nabla p + u\cdot\nabla u = f$$

$$\nabla\cdot u = 0$$

$$\underbrace{\frac{\partial c}{\partial t}}_{\text{time-dep.}} - \underbrace{\varepsilon \Delta c}_{\text{diffusion}} + \underbrace{\vec{u}\cdot\nabla c}_{\text{Convection}} + \underbrace{c\,u}_{\text{Reaction}} = f$$

Species Continuity eqn.